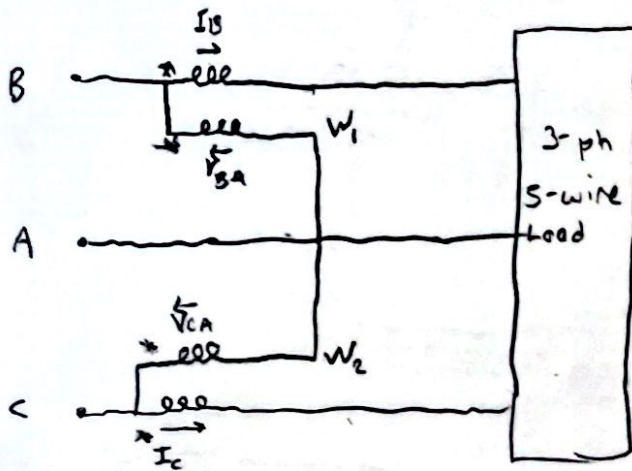


ELE 637 - Pre-Lab #1

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(1) $I_a + I_b + I_c = 0 \rightarrow I_a = -I_b - I_c$, $S = V_a I_a^* + V_b I_b^* + V_c I_c^*$

$$= V_a (-I_b^* - I_c^*) + V_b I_b^* + V_c I_c^* \\ = (-V_a + V_b) I_b^* + (-V_a + V_c) I_c^* = \boxed{V_{ba} I_b^* + V_{ca} I_c^*}$$



(2)

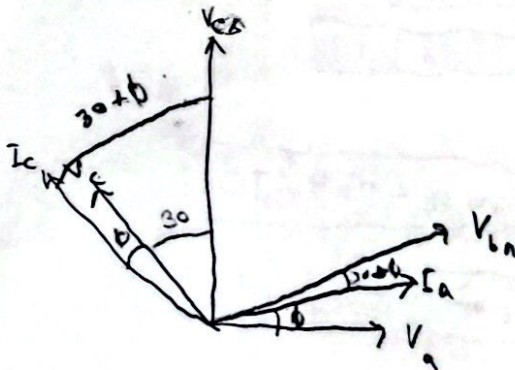
Since the 2 wattmeter method is based on a 3 wire circuit where 3 line currents $\rightarrow I_a + I_b + I_c = 0$, but 4-wire circuit $\rightarrow I_a + I_b + I_c + I_d = 0$
 $\therefore I_a + I_b + I_c \neq 0 \rightarrow \text{NA}$

RC Loads $\rightarrow W_1 = V_{AB} I_A \cos(30^\circ - \phi)$

$W_2 = V_{CB} I_C \cos(30^\circ + \phi)$

$$W_1 + W_2 = V_{LL} I_L (\cos(30^\circ - \phi) + \cos(30^\circ + \phi)) \\ = \sqrt{3} V_{LL} I_L \cos \phi \quad \checkmark$$

(3)



(4) lagging power factor load $\rightarrow \phi < 0^\circ$

$PF = \cos(\phi) < 0.5$

$W_1 \rightarrow 30^\circ + |\phi| > 90^\circ \rightarrow \text{negative}$

$W_1 = V_{AB} I_A \cos(30^\circ + \phi) \rightarrow V_{AB} I_A \cos(>90^\circ) \rightarrow \text{negative}$

(5) (a) Resistive $\rightarrow \phi = 0^\circ$

$W_1 = \frac{\sqrt{3}}{2} V_{LL} I_L$

$W_2 = \frac{\sqrt{3}}{2} V_{LL} I_L$

(b) Inductive $\rightarrow \phi = 90^\circ$

$W_1 = \frac{1}{2} V_{LL} I_L$

$W_2 = \frac{1}{2} V_{LL} I_L$

(c) Capacitive $\rightarrow \phi = -90^\circ$

$W_1 = \frac{1}{2} V_{LL} I_L$

$W_2 = -\frac{1}{2} V_{LL} I_L$

$$(6) \quad \left. \begin{aligned} W_1 &= V_{AB} I_A \cos(30^\circ - 101^\circ) \\ &= V_{LL} I_L \cos(30^\circ - 101^\circ) \end{aligned} \right\} \left. \begin{aligned} W_2 &= V_{CB} I_C \cos(30^\circ - 101^\circ) \\ &= V_{LL} I_L \cos(30^\circ - 101^\circ) \end{aligned} \right\}$$

$$\therefore W_{tot} = W_1 + W_2 = \sqrt{3} V_{LL} I_L \cos 101^\circ$$

$$Q = \sqrt{3} (W_1 - W_2) = \sqrt{3} V_{LL} I_L (\cos(30^\circ - 101^\circ) - \cos(30^\circ - 101^\circ)) = \sqrt{3} V_{LL} I_L \sin 101^\circ$$

$$\frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2} = \frac{-\sqrt{3} V_{LL} I_L \sin 101^\circ}{\sqrt{3} V_{LL} I_L \cos 101^\circ} = -\tan 101^\circ \rightarrow \tan 101^\circ = -\frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2}$$

$$(7) \quad \text{Resistive load} \rightarrow W_1 = \frac{\sqrt{2}}{2} V_{LL} I_L, \quad W_2 = \frac{\sqrt{2}}{2} V_{LL} I_L$$

positive $\rightarrow$ correct connection

(8)

$$(a) \quad R = 20 \Omega, \quad C = 47 \mu F,$$

$$Z = R - j \frac{1}{\omega C} = 20 - j \frac{1}{377 \times 47 \mu F} = 20 - j 56.44 \Omega = 59.88 \angle -70.49^\circ \Omega$$

$$I_L = \frac{69.3V}{59.88 \angle -70.49^\circ \Omega} \rightarrow I = \frac{V}{Z} \rightarrow 1.157 \angle 70.49^\circ A$$

$$P = 69.3V \times 1.157 \cos(-70.49^\circ) = 26.78 W$$

$$Q = 69.3V \times 1.157 \sin(-70.49^\circ) = -75.58 VAR$$

$$(b) \quad R = 40 \Omega, \quad C = 47 \mu F$$

$$Z = R - j \frac{1}{\omega C} = 40 - j 56.44 \Omega = 69.18 \angle -54.67^\circ \Omega$$

$$I_L = \frac{69.3V}{69.18 \angle -54.67^\circ \Omega} = 1.002 \angle 54.67^\circ A$$

$$P = 69.3 \times 1.002 \cos(-54.67^\circ) = 40.16 W$$

$$Q = 69.3 \times 1.002 \sin(-54.67^\circ) = 66.17 VAR$$

Voltage V	R Ω	C μF	Current A	P W	Q VAR
69.3	20	47	1.157 $\angle 70.49^\circ$	26.78	-75.58
69.3	40	47	1.002 $\angle 54.67^\circ$	40.16	66.17

Table 1.
Calculation Results for
Single phase loads